THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

IRINGA REGION

FORM THREE JOINT ANNUAL EXAMINATION

BIOLOGY

PROPOSED MARKING GUIDE

1.

I	Ii	Iii	Iv	V	vi	Vii	Viii	Ix	X
B	C	C	C	B	A	C	B	E	D

(10 marks 01@mark)

2.

I	Ii	Iii	Iv	V	Vi
D	H	F	B	E	C

(06marks 01@mark)

3. i. To investigate whether exhaled air contain more carbon dioxide than inhaled air (02marks)

ii. To indicate the presence of carbon dioxide by reacting and forming calcium carbonate

iii. Because it react with carbon dioxide to form calcium carbonate which is milky

$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(aq)$ (02marks)

iv. Inhaled air contains very little carbon dioxide(02marks)

(b) i. Gaseous exchange is the movement of respiratory gases across the respiratory surface. (02marks)

ii. Gaseous exchange is importance because help to remove carbon dioxide as excretory product from the animal body. (01marks)

4. (a) i. > Carbon dioxide

>oxygen

> nutrients (02marks 0.5@mark)

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> Water molecule

> Hormones

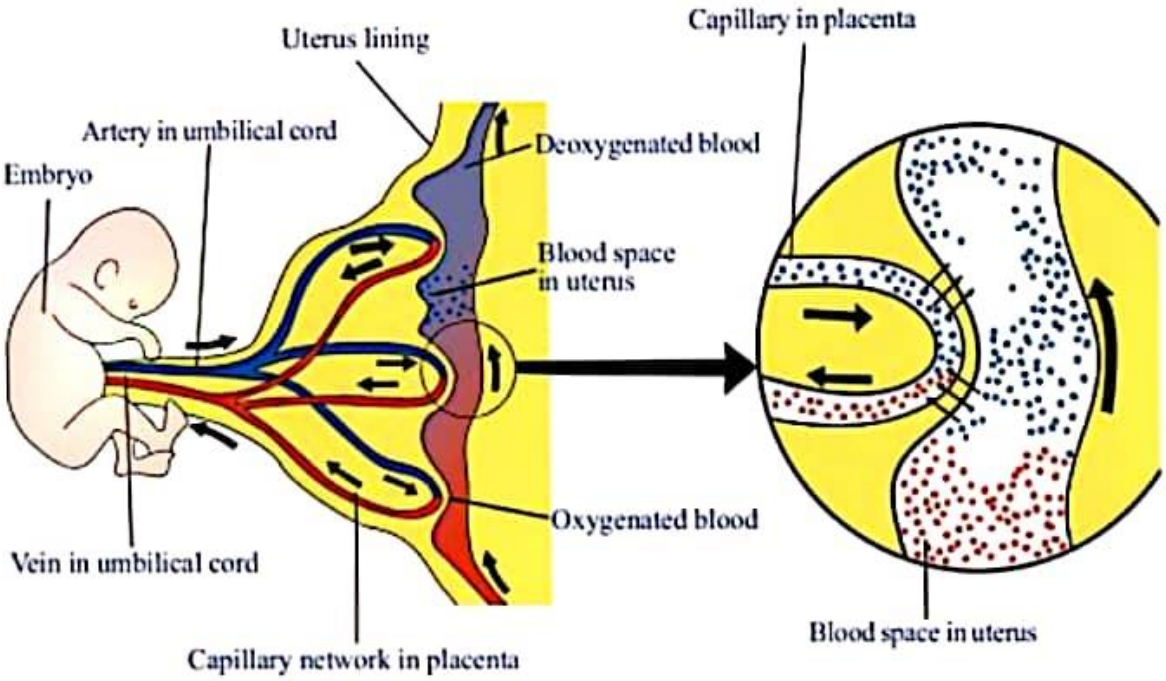
> Urea

(b) ii. Have villi which increase the surface area for diffusion of the materials

Have thin placental membrane which reduce the distance across the area of exchange.

Have blood vessel for transportation of materials exchanged. (03marks 01@mark)

(c). diagram of exchange of substance between the blood of the embryo and the mother



Caption and Labelling (05marks)

5, (a) Balanced diet refers to the meal containing foods from all food group in the right proportional, proper diet helps an individual to remain health (01mark)

(b) Water used in the digestion and absorption of food

It is a medium of transport for food and hormones

It acts as lubricants

It help in excretion of harmful by product of metabolic processes example urine and sweat.

It helps in regulation of heat loss
It used in the manufacture of different secretion example tears and saliva. (08marks 02@mark)

6 (a) Muscle cramps(02@marks)

(b) Prolonged exercise
Lack of salt in the body
Injury
Dehydration (04marks 01@mark)

(c). Stretching and warning the affected muscles
Drinking solution of salts
Gentry massaging the muscles (03marks 01@mark)

7 Data given
Total magnification = X200
Eye piece lens magnification = X20
Objective lens magnification =
Formula,

Total magnification = Objective lens magnification X Eye piece lens magnification

$$\text{Objective lens magnification} = \frac{\text{Total magnification}}{\text{Eye piece lens magnification}}$$

$$\text{Objective lens magnification} = \frac{200}{20}$$

Objective lens magnification = 10 (03marks)

(b) i. Reflect light through the condenser to the object
ii. Provide firm and steady support to the microscope
iii. Regulate the amount of light passing from the mirror to the condenser. (06marks 02@mark)

8. (a) Components of homeostatic mechanism
>Receptor ,These are the specialized cells that detect changes in the environment
>Coordinator (CNS), This is an organ which receive and interpret messages from the receptor
>Effectors . These are the cells ,organ or organelles which receive motor impulses from coordinator.
(b) (i) Regulation of water content in the body
Regulation of the pH of the body

Excretion

Regulation of the chemical composition

(ii) Deamination

Detoxification

Formation of bile

Breaks down hormones.

(iii) Excretion

Protection

Temperature regulation

Sensation.

8. (a) Viruses reproduce and carry metabolism when they are inside the living cell but when they are outside the living environment they behaves as non living (virion) (02@mark)

(b) Viruses are used by scientists in manipulation and investigation of the cell

Some viruses are used to make vaccine

Some viruses like Bacteriophage used to attack bacteria which help in controlling bacteria infection . (02marks 01@mark)

(c). Phylum Rhizopoda/Amoebozoa example amoeba

Pylum Apicomplexa example plasmodium

Phylum Euglenophyta example Euglena

Phylum ciliophera paramecium example paramecium

Phylum Zoomastigma example trypanasoma (05marks 01@mark)

SECTION C (30MARKS)

A student should attempt only two question from this section

9. (a) White blood cell

Red blood cell

Platelets

Blood plasma. (04marks 01@mark)

(b) It Transport carbon dioxide and oxygen

It aid in repairing of wound

It transport nutrients hormones and wastes

It protect the body against diseases and infections (08marks 02@mark)

(c). The wall of the left ventricle is thicker than that of the right ventricle in order to create high pressure that enable it to pump blood toward the all parts of the body. (03marks)

10. (a) Components of homeostatic mechanism

- Receptor ,These are the specialized cells that detect changes in the environment
- Coordinator (CNS), This is an organ which receive and interpret messages from the receptor
- Effectors. These are the cells, organ or organelles which receive motor impulses from coordinator. (03marks 01@mark)

(b) (i) Regulation of water content in the body

Regulation of the pH of the body

Excretion

Regulation of the chemical composition (04marks 01@mark)

(ii) Deamination

Detoxification

Formation of bile

Breaks down hormones. (04marks 01@mark)

(iii) Excretion

Protection

Temperature regulation

Sensation. (04marks 01@mark)

11. Introduction

- Meaning of cell division
- Types of cell division
- Any relevant explanation on cell division (01mark)

Main body

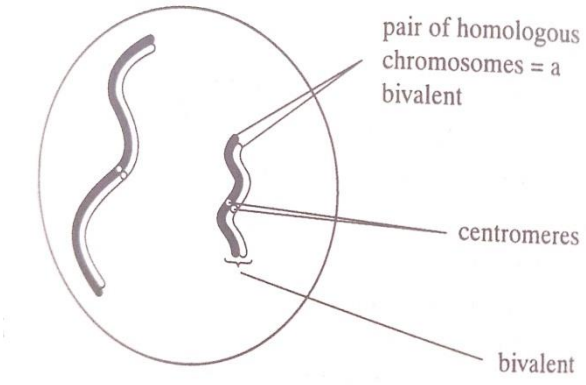
- Interphase(Resting stage) , This is the phase before meiosis begin

Events .

- ❖ DNA replication.
- ❖ Chromatins observed
- ❖ Cell grows(01mark)

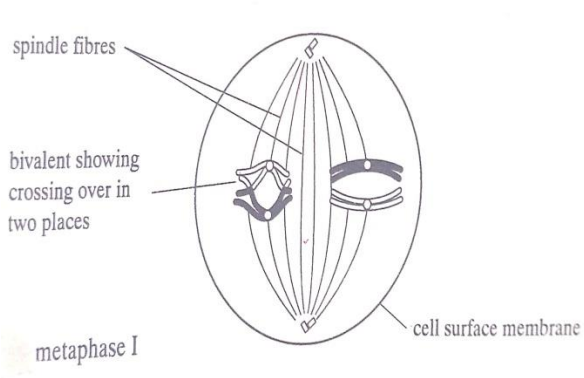
- Prophase I ,This is a longest phase of meiosis

Events

- ❖ Chromosomes become visible
- ❖ Homologous chromosomes pair up
- ❖ Crossing over occur (Any two points 0.5 @ = 01mark)
- ❖ Centrioles migrate to the pole
- ❖ Nuclear disappear
- ❖ 

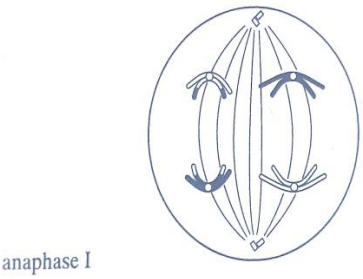
(02marks)

- Metaphase I, it involves the following events
- ❖ Spindle fibre forms
- ❖ Chromosomes are positioned at the equator (Any two points 0.5 @ = 01mark)



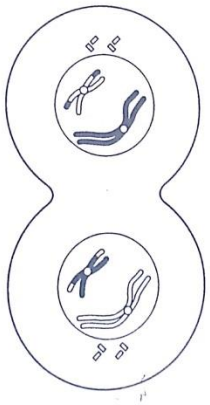
(02marks)

- Anaphase I involves
- ❖ Microtubule shorten
- ❖ Each chromosome contain the pair of sister chromatids
- ❖ Chromosomes moves toward the opposite pole (Any two points 0.5 @ = 01mark)



(02marks)

- Telophase I ,In this phase the first meiotic division effectively ends
  Events
- ❖ Each daughter cell has half number of chromosome
- ❖ Each chromosome consists pair of chromatids
- ❖ Microtubule disappear
- ❖ Cytokinesis occur in animal cell
- ❖ Chromosome uncoil back into chromatins (Any two points 0.5 @ = 01mark)



telophase I in an animal cell

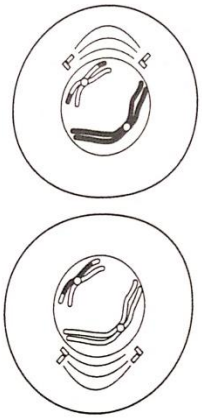
(02marks)

Since meiosis occurs in two phases , the followings are the phases for meiosis II

- Prophase II

Events

- ❖ New spindle fibres are formed at right angle
- ❖ Nuclear disintegrate
- ❖ Nuclear membrane dissolved
- ❖ Chromatids are shorten and become thicker (Any two points 0.5 @ = 01mark)



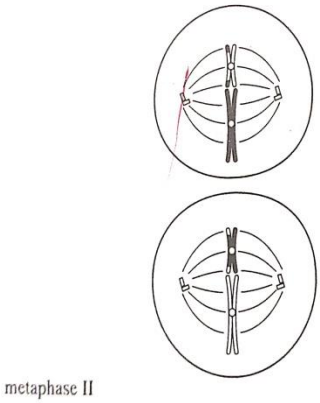
) prophase II

(02mark)

- Metaphase II

Events

- ❖ Chromosomes line up separately around the equator of the spindle fibre
- ❖ Chromatids are attached at their centromere (Any two points 0.5 @ = 01mark)



metaphase II

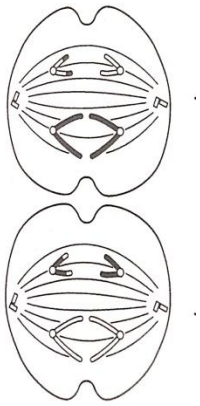
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(02marks)

- Anaphase II

  Events
- ❖ The centromeres divide and the spindle fibres pull the chromatids to opposite poles

(Any two points 0.5 @ = 01mark)



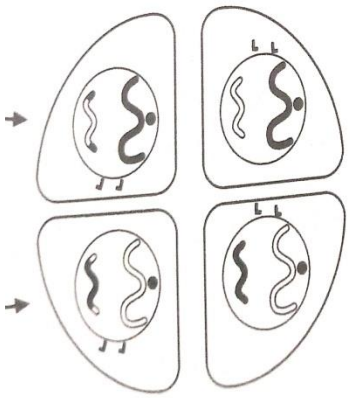
anaphase II

(02marks)

- Telophase II

  Events
- ❖ The chromosomes lengthen and become thin
- ❖ Spindle fibres disappear
- ❖ Four doughter cell are formed

(Any two points 0.5 @ = 01mark)



(02marks)

Conclusion

Any relevant conclusion (01mark)